

Session12

Part A – Setup and Data Preparation

Q1. Import the required libraries.

Q2. Create a synthetic “Employee Performance” dataset.

```
data = {  
    "Experience_Years": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10,  
                        2, 4, 6, 8, 10, 12, 14, 16, 18, 20],  
    "Training_Hours": [5, 10, 15, 20, 25, 30, 35, 40, 45, 50,  
                      8, 12, 18, 25, 35, 40, 50, 60, 65, 70],  
    "Attendance": [70, 75, 80, 82, 85, 88, 90, 92, 94, 95,  
                  74, 78, 83, 86, 89, 91, 93, 95, 97, 98],  
}
```

Part B – Model Building

Q3. Split data into training and test sets.

Q4. Train a Decision Tree Regressor.

Q5. Evaluate model performance.

Part C – Global Interpretability (SHAP)

Q6. Use SHAP to visualize which features matter most overall.

Part D – Local Interpretability (SHAP Force Plot)

Q7. Explain one specific prediction.

Part E – Local Interpretability with LIME

Q8. Create a LIME explainer for regression.

Part F – Compare SHAP vs LIME

Q9. What differences do you see between SHAP and LIME explanations?

Part G – Reflection

Q10. Why do we need interpretability methods in analytics?